Intensity, decibels and Doppler

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Learning target: Distinguish intensity change from frequency shift.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 8; 3 course days later.*
Parallel oscillation; a longitudinal pressure wave.

02. Explain a worked example (worked)
Area proportional to r^2; total power conserved while intensity decreases.

03. Look and explain (model)
Higher on approach, lower on recession in basic Doppler model.

04. Build the solution (practice)
+10 dB and +20 dB.

05. Use the idea (reason)
No; intensity/amplitude and frequency/pitch are distinct.

06. Choose a method (mixed) *Review: Unit 5, Day 11; 12 course days later.*
24°C; ΔT is proportional to absorbed energy under these conditions.

07. Exit check (exit)
Higher in basic approach model.

08. STRETCH: Apply it another way (stretch)
1, 1/4, 1/9.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 8; 3 course days later.*
150 cycles; T≈0.002 s.

02. Explain a worked example (worked)
Area proportional to r^2; total power conserved while intensity decreases.

03. Represent and interpret (model)
Higher on approach, lower on recession in basic Doppler model.

04. Solve and justify (practice)
6.02 dB; 80 dB intensity 100 times 60 dB.

05. Analyze the reasoning (reason)
No; intensity/amplitude and frequency/pitch are distinct.

06. Choose a method (mixed) *Review: Unit 5, Day 11; 12 course days later.*
Absorbed 12000 J; source 240000 J.

07. Exit check (exit)
531.25 Hz.

08. STRETCH: Apply it another way (stretch)
Intensity ratio 1/9 gives -9.54 dB. Approach compresses arriving wavefront spacing/increases observed frequency; distinct from intensity change.